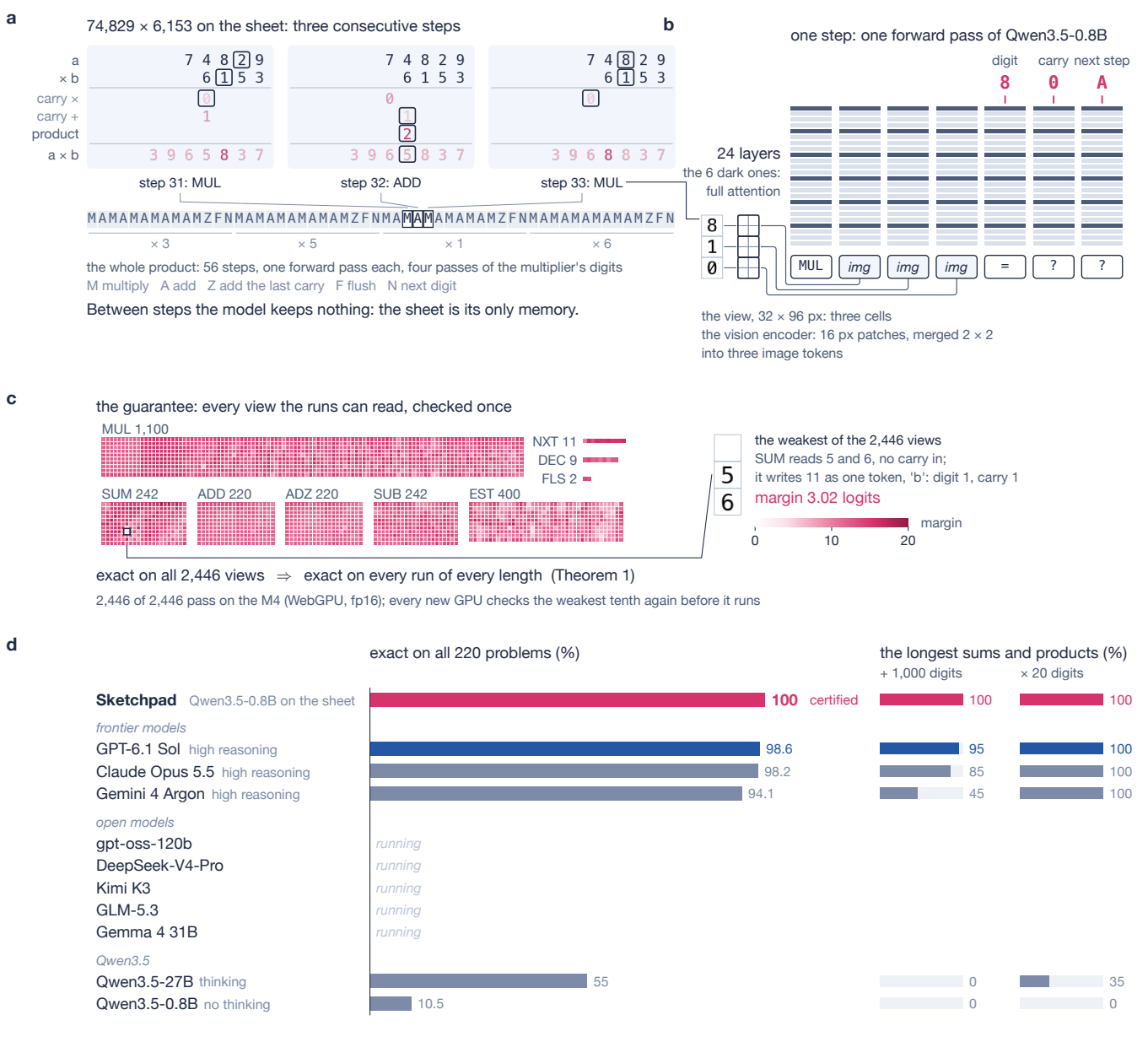

a
74,829 × 6,153 on the sheet: three consecutive steps
a
× b
carry ×
carry +
product
a × b
step 31: MUL
step 32: ADD
step 33: MUL
× 3
× 5
× 1
× 6
the whole product: 56 steps, one forward pass each, four passes of the multiplier's digits
M multiply A add Z add the last carry F flush N next digit
Between steps the model keeps nothing: the sheet is its only memory.
b
one step: one forward pass of Qwen3.5-0.8B
digit
carry
next step
24 layers
the 6 dark ones:
full attention
MUL
img
img
img
=
?
?
the view, 32 × 96 px: three cells
the vision encoder: 16 px patches, merged 2 × 2
into three image tokens
c
the guarantee: every view the runs can read, checked once
MUL 1,100
NXT 11
DEC 9
FLS 2
SUM 242
ADD 220
ADZ 220
SUB 242
EST 400
the weakest of the 2,446 views
SUM reads 5 and 6, no carry in;
it writes 11 as one token, 'b': digit 1, carry 1
margin 3.02 logits
margin
0
10
20
exact on all 2,446 views ⇒ exact on every run of every length (Theorem 1)
2,446 of 2,446 pass on the M4 (WebGPU, fp16); every new GPU checks the weakest tenth again before it runs
d
exact on all 220 problems (%)
the longest sums and products (%)
+ 1,000 digits
× 20 digits
Sketchpad Qwen3.5-0.8B on the sheet
100
certified
frontier models
GPT-6.1 Sol high reasoning
98.6
Claude Opus 5.5 high reasoning
98.2
Gemini 4 Argon high reasoning
94.1
open models
gpt-oss-120b
DeepSeek-V4-Pro
Kimi K3
GLM-5.3
Gemma 4 31B
running
Qwen3.5
Qwen3.5-27B thinking
55
Qwen3.5-0.8B no thinking
10.5